

Building the Analytic Spine — Navrongo HDSS Extract

Heat Stress and Perinatal Outcomes in Ghana

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1 Purpose and scope

This document implements steps (ii)–(iv) of the analytic plan accompanying the proposal *Heat Stress and Perinatal Outcomes in Ghana: A Spatio-Temporal Analysis* (Version 2.0). It produces the cleaned analytic spine that all downstream modelling scripts will consume; it does not itself fit any statistical model.

The three operations executed here are:

- **(ii) Deduplication.** The raw extract contains one row per maternal residency episode, which yields multiple rows per child when a mother’s residency history spans more than one episode. The spine is reduced to one row per unique child, and the maternal residency episodes are retained in a separate long-format table that downstream survival or migration-history models can join back to.

- **(iii) Recoding of HDSS missing-data sentinels.** Per the NHRC value-label codebook (delivered 21 May 2026), the integers 8 and 9 denote “NaN/missing” and “DK” in the categorical fields. The codebook conventions are applied here. In addition, earlier correspondence flagged 88 and 99 as missing in `ceb`, `prgmonth`, and `numanc`; these are also recoded defensively, though neither appears in the present sample. The value 0 in `child_bwgt` denotes a child who was not weighed; biologically implausible weights (>6 kg or <0.5 kg) and the residual code 9.9 are also set to NA. Blank cells in `mother_mstatus` are recoded to NA.
- **(iv) Construction of derived analytic variables.** Mortality indicators (stillbirth-as-recorded, day-0, early-neonatal, neonatal, infant, under-five), birth-weight categories (LBW, VLBW), maternal-age and parity categories, season of delivery on both the three-season and two-season conventions for northern Ghana, reconstructed pregnancy windows for downstream DLNM linkage, and a flag for the **post-2012 ANC-reliable sub-cohort** used in the care-modifier sub-analyses.

Variable name reconciliation. The 21 May 2026 sample uses three field names that differ from the 30 April 2026 sample: `cordres1` $\rightarrow$ `corddress`, `delivery` $\rightarrow$ `delivery_type`, `gender` $\rightarrow$ `child_gender`. The script uses the new names throughout; if it is re-run against the older extract these will need to be renamed back.

Analytic window. Following the workflow agreed with NHRC, the primary analytic window covers *all available birth-outcome records* in the full extract, rather than a fixed ten-year span. This decision reflects the relative scarcity of perinatal-mortality events and the value of preserving the full longitudinal record for the main outcome models. The antenatal-care variables (`numanc`, `prgmonth`, `ancatt`) are reliably populated only from 2013 onward; analyses that adjust for or stratify on ANC are therefore restricted to a 2013-onward sub-cohort, with a binary flag (`anc_reliable`) constructed in the spine for that purpose.

Geocodes. Household-level coordinates (`gps_latitude`, `gps_longitude`) are present in this sample. Coverage is partial in the sample (concentrated in more recent years); the full extract is expected to have broader coverage. The spine preserves the coordinates as-is for downstream linkage in `02_link_era5_utci.Rmd`.

The numeric category codes for `assist_del`, `child_birth_place`, `delivery_type`, `cutcod`, `corddress`, `seventtype`, and `eeventtype` will be converted to labelled factors in a follow-up labelling chunk using the NHRC codebook delivered on 20 May 2026.

2 Setup

```
# Install any missing packages on first run:
# install.packages(c("haven", "dplyr", "tidyr", "lubridate", "stringr",
#                   "readr", "knitr", "kableExtra"))
library(haven)      # read_dta
library(dplyr)
library(tidyr)
library(lubridate)
library(stringr)
library(readr)
library(knitr)
library(kableExtra)

# Adjust these paths to match the project layout.
IN_PATH <- "vincent_phd_10data_20052026.dta"
OUT_DIR <- "."
dir.create(OUT_DIR, showWarnings = FALSE, recursive = TRUE)
```

A small helper used throughout the document to render diagnostic tables in a consistent style:

```
nice_table <- function(x, caption = NULL, full_width = FALSE) {
  kable(x, caption = caption, format.args = list(big.mark = ",")) |>
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = full_width, position = "left")
}
```

3 Load raw data

`haven::read_dta()` preserves the Stata variable labels and value labels as attributes, so they remain accessible for documentation while we operate on the underlying numeric codes.

```
raw <- read_dta(IN_PATH)

tibble(metric = c("rows", "columns",
  "unique mothers (motherid)",
  "unique children (child_id)",
  "unique compounds (locationid)"),
  value = c(nrow(raw), ncol(raw),
    dplyr::n_distinct(raw$motherid),
    dplyr::n_distinct(raw$child_id),
    dplyr::n_distinct(raw$locationid))) |>
  nice_table(caption = "Raw extract - basic dimensions")
```

Table 1: Raw extract — basic dimensions

metric	value
rows	22,340
columns	29
unique mothers (motherid)	5,526
unique children (child_id)	8,876
unique compounds (locationid)	3,053

The difference between raw rows and unique children reflects the residency-episode duplication that step (ii) will resolve.

4 Step (iii): Recode HDSS missing-data sentinels

Per the NHRC value-label codebook of 20 May 2026, the integers 8 (NaN) and 9 (DK) denote missing values across the categorical fields. The values 88 and 99 were flagged in earlier correspondence as missing in `ceb`, `prgmonth`, and `numanc` and are recoded defensively, though neither appears in the present sample. The value 0 in `child_bwgt` denotes a child who was not weighed; biologically implausible weights and the residual code 9.9 are also set to NA.

```
sentinel_cat <- c(8, 9)
sentinel_num <- c(88, 99)

df <- raw |>
  mutate(
    ceb = if_else(ceb %in% sentinel_num, NA_real_, ceb),
    prgmonth = if_else(prgmonth %in% sentinel_num, NA_real_, prgmonth),
    numanc = if_else(numanc %in% sentinel_num, NA_real_, numanc),
    across(c(assist_del, child_birth_place, delivery_type, cutcod,
```

```

        corddress, ancatt, first_nb, child_birth_type),
    ~ if_else(.x %in% sentinel_cat, NA_real_, .x)),
child_bwgt    = if_else(child_bwgt == 0 |
                        child_bwgt == 9.9 |
                        child_bwgt < 0.5 |
                        child_bwgt > 6.0,
                        NA_real_, child_bwgt),
mother_mstatus = as.character(mother_mstatus),
mother_mstatus = if_else(str_trim(mother_mstatus) == "",
                        NA_character_, mother_mstatus)
)

```

Effect of the recode on the non-missing counts of the affected fields:

```

before_vals <- c(
  ceb = sum(!is.na(raw$ceb)),
  prgmonth = sum(!is.na(raw$prgmonth)),
  numanc = sum(!is.na(raw$numanc)),
  child_bwgt = sum(!is.na(raw$child_bwgt)),
  mother_mstatus = sum(!is.na(as.character(raw$mother_mstatus)) &
                        trimws(as.character(raw$mother_mstatus)) != "")
)

after_vals <- c(
  ceb = sum(!is.na(df$ceb)),
  prgmonth = sum(!is.na(df$prgmonth)),
  numanc = sum(!is.na(df$numanc)),
  child_bwgt = sum(!is.na(df$child_bwgt)),
  mother_mstatus = sum(!is.na(df$mother_mstatus))
)

tibble(
  variable = names(before_vals),
  before_n = as.integer(before_vals),
  after_n = as.integer(after_vals),
  n_recoded_NA = before_n - after_n
) |>
  nice_table(caption = "Non-missing counts before and after sentinel recoding")

```

Table 2: Non-missing counts before and after sentinel recoding

variable	before_n	after_n	n_recoded_NA
ceb	12,360	12,360	0
prgmonth	22,184	22,184	0
numanc	22,158	22,158	0
child_bwgt	6,442	6,423	19
mother_mstatus	22,337	22,337	0

5 Step (ii.a): Build the residency-episode table

Each maternal residency episode is kept on its own row in a separate long-format table. This file is the join key for any analysis that needs in-migration or out-migration timing.

```

residency_episodes <- df |>
  distinct(motherid, sdate, edate, seventtype, eeventtype,
           locationid, location) |>
  arrange(motherid, sdate) |>
  mutate(episode_id = row_number(),
         sdate = as.Date(sdate),
         edate = as.Date(edate))

tibble(metric = c("episode rows", "unique mothers represented"),
       value = c(nrow(residency_episodes),
                  dplyr::n_distinct(residency_episodes$motherid))) |>
  nice_table(caption = "Residency-episode table - dimensions")

```

Table 3: Residency-episode table — dimensions

metric	value
episode rows	15,406
unique mothers represented	5,526

A small preview of the table (sorted by mother and episode start date), to confirm that the structure is as expected:

```

head(residency_episodes, 6) |>
  nice_table(caption = "First six rows of the residency-episode table")

```

Table 4: First six rows of the residency-episode table

motherid	sdate	edate	seventtype	eeventtype	locationid	location	episode_id
CAA001011	1995-11-16	1996-11-15	3	2	CAU004	2	1
CAA001011	2001-12-14	2004-04-25	2	2	CAU004	2	2
CAA001011	2004-04-25	2005-06-28	2	2	CAU004	2	3
CAA001011	2006-11-15	2010-04-15	2	2	CAU004	2	4
CAA001011	2011-07-15	2015-07-15	2	1	CAU004	2	5
CAA001013	1995-11-16	2016-07-15	3	2	CAA001	2	6

6 Step (ii.b): Deduplicate to one row per child

Where multiple rows share a `child_id`, the surviving row is chosen by three ordered criteria. The first is **outcome preservation**: rows with a definite birth outcome (`child_birth_type` = 1–5, covering abortion, miscarriage, live birth, stillbirth, and neonatal death) take precedence over the pregnancy-still-in-progress codes (6 = censored preg, 7 = still preg) and over missing/DK codes. This avoids the failure mode in which stillbirths and very-early neonatal deaths recorded at the time of the pregnancy event are overwritten by later residency-only rows that carry a “censored pregnancy” code. The second is lowest count of missing values across the analytic covariate set. The third is tie-break on the most recent `sdate`.

```

analytic_vars <- c("mother_educ", "ses_index", "mother_mstatus",
                  "child_bwgt", "child_birth_place", "delivery_type",
                  "ancatt", "prgmonth", "numanc",
                  "first_nb", "assist_del", "cutcod", "corddress")

# Priority for outcome codes: definite outcomes (1-5) win over pregnancy

```

```

# in progress (6,7) which win over missing (NA).
outcome_priority <- function(x) {
  case_when(
    x %in% c(1, 2, 3, 4, 5) ~ 1L, # abortion, miscarriage, live birth, SB, ND
    x == 6 ~ 2L, # censored preg
    x == 7 ~ 3L, # still preg
    TRUE ~ 9L # missing/DK
  )
}

spine_pre <- df |>
  mutate(.outcome_priority = outcome_priority(child_birth_type),
         .miss_score = rowSums(is.na(across(all_of(analytic_vars))))) |>
  arrange(child_id, .outcome_priority, .miss_score, desc(sdate)) |>
  distinct(child_id, .keep_all = TRUE) |>
  dplyr::select(-.outcome_priority, -.miss_score)

stopifnot(!anyDuplicated(spine_pre$child_id))

tibble(metric = c("rows in spine (= unique children)",
                  "rows in raw extract",
                  "rows removed by deduplication"),
       value = c(nrow(spine_pre),
                  nrow(df),
                  nrow(df) - nrow(spine_pre))) |>
  nice_table(caption = "Effect of child-level deduplication")

```

Table 5: Effect of child-level deduplication

metric	value
rows in spine (= unique children)	8,876
rows in raw extract	22,340
rows removed by deduplication	13,464

7 Step (iv): Derived analytic variables

Several conventions used in this block deserve explicit mention.

- Age at death is computed as integer days from `child_dob` to `child_dod`. Children without a recorded death date are coded as NA for age at death, and as 0 (not NA) for the binary mortality indicators, so that those indicators are NA-safe for downstream regression.
- The 280-day term-gestation assumption used to back-date pregnancy start is a working default. Sensitivity analyses should also fit results at 259 and 294 days to bracket gestational-age error.
- Northern Ghana climatology follows the conventional three-season split: *dry-cool* (Nov–Mar), *hot-dry* (Apr–May), and *wet* (Jun–Oct). A two-level wet-vs-dry version is also retained for parsimonious models.
- Parity is taken from `ceb` where available. Because `ceb` is missing for a substantial fraction of records (approximately 45% in the 20 May 2026 sample), a binary nulliparous proxy is also computed from the fully-populated `first_nb` field.

```

spine <- spine_pre |>
  mutate(
    # ---- Dates ----

```

```

mother_dob = as.Date(mother_dob),
child_dob  = as.Date(child_dob),
child_dod  = as.Date(child_dod),
sdate      = as.Date(sdate),
edate      = as.Date(edate),

# ---- Age at death (days) ----
age_death_days = as.integer(child_dod - child_dob),

# ---- Mortality indicators (NA-safe) ----
died_day0      = if_else(!is.na(age_death_days) & age_death_days == 0,      1L, 0L),
died_enh       = if_else(!is.na(age_death_days) & age_death_days >= 0
                        & age_death_days <= 7,                             1L, 0L),
died_neonatal  = if_else(!is.na(age_death_days) & age_death_days >= 0
                        & age_death_days <= 28,                           1L, 0L),
died_infant    = if_else(!is.na(age_death_days) & age_death_days >= 0
                        & age_death_days <= 365,                          1L, 0L),
died_u5       = if_else(!is.na(age_death_days) & age_death_days >= 0
                        & age_death_days <= 1826,                         1L, 0L),

# ---- Stream A primary outcome: stillbirth + ENN composite ----
# Stillbirth as recorded in child_birth_type (code = 4)
stillbirth = if_else(!is.na(child_birth_type) & child_birth_type == 4, 1L, 0L),
# Perinatal mortality (Stream A primary): stillbirth OR death within
# the first 7 days of life
perinatal  = if_else(stillbirth == 1 | died_enh == 1, 1L, 0L),

# ---- Birth-weight derivatives ----
lbw = if_else(!is.na(child_bwgt), as.integer(child_bwgt < 2.5), NA_integer_),
vlbw = if_else(!is.na(child_bwgt), as.integer(child_bwgt < 1.5), NA_integer_),

# ---- Temporal markers ----
birth_year = year(child_dob),
birth_month = month(child_dob),
season3 = case_when(
  birth_month %in% c(11, 12, 1, 2, 3) ~ "dry_cool",
  birth_month %in% c(4, 5)           ~ "hot_dry",
  birth_month %in% c(6, 7, 8, 9, 10) ~ "wet",
  TRUE                             ~ NA_character_),
season2 = if_else(birth_month %in% 6:10, "wet", "dry"),

# ---- Maternal age and parity ----
mat_age_yrs = as.numeric(child_dob - mother_dob) / 365.25,
mat_age_cat = cut(mat_age_yrs,
  breaks = c(-Inf, 19.999, 24, 29, 34, Inf),
  labels = c("<20", "20-24", "25-29", "30-34", "35+"),
  right = TRUE),
parity_cat = cut(ceb,
  breaks = c(0, 1, 2, 4, Inf),
  labels = c("nulliparous", "1", "2-4", "5+"),
  include.lowest = TRUE, right = TRUE),
nulliparous_proxy = if_else(first_nb == 1, 1L, 0L),

```

```

# ---- Education collapsed ----
educ_any_secondary = if_else(!is.na(mother_educ),
                             as.integer(mother_educ >= 2),
                             NA_integer_),

# ---- Reconstructed pregnancy windows (280-day term gestation) ----
preg_start_assumed = child_dob - days(280),
preg_t3_start      = child_dob - days(98),
preg_last30        = child_dob - days(30),
preg_last7         = child_dob - days(7),

# ---- ANC-reliable sub-cohort flag ----
# The antenatal-care fields are populated reliably only for births from
# 2013 onward; analyses that adjust for or stratify on ANC are restricted
# to this sub-cohort. The PRIMARY analytic window uses all available
# birth-outcome records (no upper or lower year restriction).
anc_reliable = as.integer(birth_year >= 2013),

# ---- Labelled birth outcome ----
outcome_class = case_when(
  child_birth_type == 1 ~ "abortion",
  child_birth_type == 2 ~ "miscarriage",
  child_birth_type == 3 ~ "live_birth",
  child_birth_type == 4 ~ "stillbirth",
  child_birth_type == 5 ~ "neonatal_death_at_record",
  child_birth_type == 6 ~ "censored_pregnancy",
  child_birth_type == 7 ~ "ongoing_pregnancy",
  TRUE ~ NA_character_)
)

```

8 Diagnostics

8.1 Mortality event counts

The full-file event counts, before any analytic-window restriction:

```

spine |>
  summarise(
    across(c(died_day0, died_enn, died_neonatal, died_infant, died_u5), sum)
  ) |>
  pivot_longer(everything(), names_to = "indicator", values_to = "events") |>
  mutate(definition = recode(indicator,
    died_day0      = "Day-0 death (death on day of birth)",
    died_enn       = "Early-neonatal death (0-7 days)",
    died_neonatal  = "Neonatal death (0-28 days)",
    died_infant    = "Infant death (0-365 days)",
    died_u5       = "Under-five death (0-1,826 days)"
  )) |>
  dplyr::select(indicator, definition, events) |>
  nice_table(caption = "Mortality event counts, full file (1984-2023)")

```

Table 6: Mortality event counts, full file (1984–2023)

indicator	definition	events
died_day0	Day-0 death (death on day of birth)	44
died_enn	Early-neonatal death (0–7 days)	122
died_neonatal	Neonatal death (0–28 days)	173
died_infant	Infant death (0–365 days)	413
died_u5	Under-five death (0–1,826 days)	602

8.2 ANC-reliable sub-cohort (2013 onward)

The 2013-onward sub-cohort is the operative one for any analysis that requires the ANC variables, since these fields are systematically blank for births before 2012. The full analytic window — used for the primary perinatal outcome models — places no upper or lower year restriction beyond what is present in the full extract delivered by NHRC.

```
sub <- spine |> filter(anc_reliable == 1)

tibble(
  metric = c("n children in ANC-reliable sub-cohort",
    "early-neonatal deaths (0-7 days)",
    "neonatal deaths (0-28 days)",
    "infant deaths (0-365 days)",
    "records with valid birth weight",
    "low birth weight (<2.5 kg)",
    "records with non-missing numanc",
    "records with valid parity (ceb)"),
  value = c(nrow(sub),
    sum(sub$died_enn,      na.rm = TRUE),
    sum(sub$died_neonatal, na.rm = TRUE),
    sum(sub$died_infant,   na.rm = TRUE),
    sum(!is.na(sub$child_bwgt)),
    sum(sub$lbw,          na.rm = TRUE),
    sum(!is.na(sub$numanc)),
    sum(!is.na(sub$ceb)))
) |>
  nice_table(caption = "Event and observation counts in the ANC-reliable (2013-onward) sub-cohort")
```

Table 7: Event and observation counts in the ANC-reliable (2013-onward) sub-cohort

metric	value
n children in ANC-reliable sub-cohort	2,942
early-neonatal deaths (0–7 days)	19
neonatal deaths (0–28 days)	26
infant deaths (0–365 days)	52
records with valid birth weight	2,059
low birth weight (<2.5 kg)	225
records with non-missing numanc	2,941
records with valid parity (ceb)	1,963

8.3 Distribution checks

Season of delivery in the ANC-reliable (2013-onward) sub-cohort, on both the three-season and the two-season conventions:

```
sub |>
  count(season3, name = "n") |>
  arrange(desc(n)) |>
  nice_table(caption = "Season of delivery (three-season), ANC-reliable sub-cohort")
```

8: Season of delivery (three-season), ANC-reliable sub-cohort

season3	n
wet	1,301
dry_cool	1,145
hot_dry	496

```
sub |>
  count(season2, name = "n") |>
  arrange(desc(n)) |>
  nice_table(caption = "Season of delivery (wet/dry), ANC-reliable sub-cohort")
```

9: Season of delivery (wet/dry), ANC-reliable sub-cohort

season2	n
dry	1,641
wet	1,301

Maternal age category, full file (a useful background distribution to keep in mind when interpreting age-stratified outputs later):

```
spine |>
  count(mat_age_cat, name = "n") |>
  arrange(mat_age_cat) |>
  nice_table(caption = "Maternal age category, full file")
```

Table 10: Maternal age category, full file

mat_age_cat	n
<20	771
20-24	1,467
25-29	2,437
30-34	2,125
35+	2,076

9 Spine codebook

The variable list below documents the derived analytic spine. It is also saved to `spine_codebook.csv` for reference.

```
spine_codebook <- tribble(
  ~variable,      ~description,
  "child_id",     "Unique child identifier (NHRC).",
  "motherid",     "Unique mother identifier.",
  "locationid",   "Compound identifier.",
```

```

"gps_latitude",      "Household-level GPS latitude (decimal degrees).",
"gps_longitude",    "Household-level GPS longitude (decimal degrees).",
"location",         "Rural/Urban indicator (1=rural, 2=urban).",
"child_dob",        "Date of birth (delivery).",
"child_dod",        "Date of death; NA if alive at last record.",
"age_death_days",   "Integer days from child_dob to child_dod; NA if alive.",
"died_day0",        "1 if death on day of birth.",
"died_enn",         "1 if death within 0-7 days of birth (early neonatal).",
"died_neonatal",    "1 if death within 0-28 days (neonatal).",
"died_infant",      "1 if death within 0-365 days (infant).",
"died_u5",          "1 if death within 0-1,826 days (under-five).",
"stillbirth",       "1 if child_birth_type = 4 (stillbirth as recorded).",
"perinatal",        "Stream A primary outcome: 1 if stillbirth OR died_enn.",
"child_gender",     "Sex of child (1=male, 2=female; codebook applies).",
"child_bwgt",       "Birth weight in kg; 0, 9.9, and outliers set to NA.",
"lbw",             "1 if birth weight < 2.5 kg; NA if weight missing.",
"vlbw",            "1 if birth weight < 1.5 kg; NA if weight missing.",
"child_birth_place", "Place of birth (1=hospital, 2=health centre, 3=home, 4=tba home, 5=other; NHRC codebook).",
"delivery_type",    "Mode of delivery (1=normal vaginal, 2=cesarean, 3=vacuum, 4=forceps, 5=vaginal).",
"assist_del",       "Delivery assistance (1=doctor, 2=midwife, 3=nurse, 4=CHO, 5=TBA, 6=family, 7=other).",
"cutcod",          "Material used to cut umbilical cord (1=new razor, 2=old razor, 3=knife, 4=sharp object).",
"corddress",       "Material used to dress umbilical stump (1=shea butter, 2=black powder + shea, 3=other).",
"ancatt",          "ANC attendance (1=yes, 2=no); reliable only 2013+.",
"prgmonth",        "Pregnancy month at first ANC visit; reliable only 2013+.",
"numanc",          "Number of ANC visits during pregnancy; reliable only 2013+.",
"first_nb",        "1 if first live birth of mother; 2 otherwise.",
"ceb",             "Children ever born.",
"parity_cat",      "Parity categories from ceb; NA where ceb missing.",
>nulliparous_proxy", "1 if first_nb==1 (use when ceb missing).",
"mother_dob",      "Mother date of birth.",
"mat_age_yrs",     "Maternal age at delivery (years).",
"mat_age_cat",     "Maternal-age categories: <20, 20-24, 25-29, 30-34, 35+.",
"mother_educ",     "Maternal education (0=none, 1=primary, 2=JHS, 3=SHS, 4=tertiary; NHRC codebook).",
"educ_any_secondary", "1 if mother_educ >= 2 (JHS or higher).",
"mother_mstatus",  "Marital status (1=divorced, 2=died, 3=married, 4=separated, 5=single, 6=widowed).",
"ses_index",       "SES quintile (1=poorest - 5=least poor; NHRC codebook).",
"birth_year",      "Calendar year of birth.",
"birth_month",     "Calendar month of birth.",
"season3",         "Season of delivery (dry_cool / hot_dry / wet).",
"season2",         "Season of delivery (dry / wet).",
"preg_start_assumed", "Assumed pregnancy start (child_dob - 280 days).",
"preg_t3_start",   "Start of third trimester (child_dob - 98 days).",
"preg_last30",     "30 days before delivery.",
"preg_last7",      "7 days before delivery.",
"anc_reliable",    "1 if birth in 2013 onward (ANC-reliable sub-cohort).",
"outcome_class",   "Labelled child_birth_type (abortion / miscarriage / live_birth / stillbirth / stillbirth)
)

nice_table(spine_codebook, caption = "Spine codebook")

```

Table 11: Spine codebook

variable	description
----------	-------------

child_id	Unique child identifier (NHRC).
motherid	Unique mother identifier.
locationid	Compound identifier.
gps_latitude	Household-level GPS latitude (decimal degrees).
gps_longitude	Household-level GPS longitude (decimal degrees).
location	Rural/Urban indicator (1=rural, 2=urban).
child_dob	Date of birth (delivery).
child_dod	Date of death; NA if alive at last record.
age_death_days	Integer days from child_dob to child_dod; NA if alive.
died_day0	1 if death on day of birth.
died_enn	1 if death within 0–7 days of birth (early neonatal).
died_neonatal	1 if death within 0–28 days (neonatal).
died_infant	1 if death within 0–365 days (infant).
died_u5	1 if death within 0–1,826 days (under-five).
stillbirth	1 if child_birth_type = 4 (stillbirth as recorded).
perinatal	Stream A primary outcome: 1 if stillbirth OR died_enn.
child_gender	Sex of child (1=male, 2=female; codebook applies).
child_bwtg	Birth weight in kg; 0, 9.9, and outliers set to NA.
lbw	1 if birth weight < 2.5 kg; NA if weight missing.
vlbw	1 if birth weight < 1.5 kg; NA if weight missing.
child_birth_place	Place of birth (1=hospital, 2=health centre, 3=home, 4=tba home, 5=other; NHRC codebook).
delivery_type	Mode of delivery (1=normal vaginal, 2=cesarean, 3=vacuum, 4=forceps, 5=vaginal cutting, 7=other; NHRC codebook).
assist_del	Delivery assistance (1=doctor, 2=midwife, 3=nurse, 4=CHO, 5=TBA, 6=family, 7=other; NHRC codebook).
cutcod	Material used to cut umbilical cord (1=new razor, 2=old razor, 3=knife, 4=sharp stone, 5=scissors; NHRC codebook).
corddress	Material used to dress umbilical stump (1=shea butter, 2=black powder + shea, 3=western medicine; NHRC codebook).
ancatt	ANC attendance (1=yes, 2=no); reliable only 2013+.
prgmonth	Pregnancy month at first ANC visit; reliable only 2013+.
numanc	Number of ANC visits during pregnancy; reliable only 2013+.
first_nb	1 if first live birth of mother; 2 otherwise.
ceb	Children ever born.
parity_cat	Parity categories from ceb; NA where ceb missing.
nulliparous_proxy	1 if first_nb==1 (use when ceb missing).
mother_dob	Mother date of birth.
mat_age_yrs	Maternal age at delivery (years).
mat_age_cat	Maternal-age categories: <20, 20-24, 25-29, 30-34, 35+.
mother_educ	Maternal education (0=none, 1=primary, 2=JHS, 3=SHS, 4=tertiary; NHRC codebook).
educ_any_secondary	1 if mother_educ >= 2 (JHS or higher).
mother_mstatus	Marital status (1=divorced, 2=died, 3=married, 4=separated, 5=single, 6=widowed; NHRC codebook).
ses_index	SES quintile (1=poorest – 5=least poor; NHRC codebook).
birth_year	Calendar year of birth.
birth_month	Calendar month of birth.
season3	Season of delivery (dry_cool / hot_dry / wet).
season2	Season of delivery (dry / wet).
preg_start_assumed	Assumed pregnancy start (child_dob - 280 days).
preg_t3_start	Start of third trimester (child_dob - 98 days).
preg_last30	30 days before delivery.
preg_last7	7 days before delivery.
anc_reliable	1 if birth in 2013 onward (ANC-reliable sub-cohort).
outcome_class	Labelled child_birth_type (abortion / miscarriage / live_birth / stillbirth / neonatal_death_at_

10 Persist outputs

The spine and the residency-episode table are written in both R-native (.rds) and text (.csv) formats so that downstream scripts in R, Stata, Python, or any other environment can read them without modification.

```
saveRDS(spine, file.path(OUT_DIR, "spine.rds"))
saveRDS(residency_episodes, file.path(OUT_DIR, "residency_episodes.rds"))
write_csv(spine, file.path(OUT_DIR, "spine.csv"))
write_csv(residency_episodes, file.path(OUT_DIR, "residency_episodes.csv"))
write_csv(spine_codebook, file.path(OUT_DIR, "spine_codebook.csv"))

cat("Files written to ", normalizePath(OUT_DIR), ":\n ",
    paste(c("spine.rds", "residency_episodes.rds",
            "spine.csv", "residency_episodes.csv",
            "spine_codebook.csv"),
          collapse = "\n "), sep = "")
```

```
#> Files written to C:\Users\bedia\Downloads\Chapter 3_Dissertation:
#>  spine.rds
#>  residency_episodes.rds
#>  spine.csv
#>  residency_episodes.csv
#>  spine_codebook.csv
```

11 Expected diagnostic counts (sample dataset, 20 May 2026)

The script has been validated against the second sample extract delivered on 21 May 2026 (22,340 rows out of approximately 223,440 in the full extract, covering birth outcomes and maternal residency from 1995 to 2020). A successful run on that sample should reproduce the following counts; any discrepancy indicates that something has changed in the input file and should be investigated before downstream modelling proceeds. The full-data run, performed by the Centre's analysts under the agreed workflow, will produce substantially larger figures; the expected-counts table is preserved as a sample-stage validation reference, not as a target for the full data.

Table 12: Expected diagnostic counts for the 20 MAY 2026 SAMPLE. Full-data figures will differ.

metric	expected_value
rows in raw extract	22,340
unique children (spine)	8,876
unique mothers	5,025
unique compounds (locationid)	2,885
distinct household GPS coordinates	904
records with non-missing GPS	2,696 (30%)
birth-year range (sample)	1995–2020
recorded stillbirths (child_birth_type = 4)	1
early-neonatal deaths (0–7 days, full file)	122
neonatal deaths (0–28 days, full file)	173
perinatal mortality composite (SB + ENN)	123
n children in ANC-reliable sub-cohort (2013+)	2,942
valid birth weights (0.5–6.0 kg)	3,204
LBW (<2.5 kg)	347

12 Session information

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14 ucrt)
#> Platform: x86_64-w64-mingw32/x64
#> Running under: Windows 11 x64 (build 26200)
#>
#> Matrix products: default
#>
#>
#> locale:
#> [1] LC_COLLATE=English_United States.utf8
#> [2] LC_CTYPE=English_United States.utf8
#> [3] LC_MONETARY=English_United States.utf8
#> [4] LC_NUMERIC=C
#> [5] LC_TIME=English_United States.utf8
#>
#> time zone: America/New_York
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods    base
#>
#> other attached packages:
#> [1] kableExtra_1.4.0 knitr_1.50      readr_2.1.6      stringr_1.6.0
#> [5] lubridate_1.9.4  tidyr_1.3.2     dplyr_1.2.0      haven_2.5.5
#>
#> loaded via a namespace (and not attached):
#> [1] bit_4.6.0          crayon_1.5.3      compiler_4.4.1    tidyselect_1.2.1
#> [5] xml2_1.5.1         parallel_4.4.1    textshaping_1.0.4 systemfonts_1.3.1
#> [9] scales_1.4.0       yaml_2.3.11       fastmap_1.2.0     R6_2.6.1
#> [13] generics_0.1.4     forcats_1.0.1     tibble_3.3.1      svglite_2.2.2
#> [17] pillar_1.11.1      RColorBrewer_1.1-3 tzdb_0.5.0        rlang_1.1.7
#> [21] stringi_1.8.7      xfun_0.54         bit64_4.6.0-1     viridisLite_0.4.3
#> [25] timechange_0.3.0   cli_3.6.5         withr_3.0.2       magrittr_2.0.4
#> [29] digest_0.6.39      vroom_1.6.7       rstudioapi_0.17.1 hms_1.1.4
#> [33] lifecycle_1.0.5    vctrs_0.7.2       evaluate_1.0.5    glue_1.8.0
#> [37] farver_2.1.2       rmarkdown_2.30    purrr_1.2.1       tools_4.4.1
#> [41] pkgconfig_2.0.3    htmltools_0.5.8.1
```

13 Next steps

With the spine in hand, the immediate items are, in order:

1. **Apply the NHRC value-label codebook** (received 20 May 2026) for `assist_del`, `child_birth_place`, `delivery_type`, `cutcod`, `corddress`, `mother_mstatus`, `child_gender`, `location`, `ses_index`, `mother_educ`, `child_birth_type`, `seventtype`, and `eeventtype`, and add a labelling chunk to this document that promotes these integer fields to labelled factors.
2. Write `02_link_era5_utci.Rmd`, which (i) reads `spine.rds`,
(ii) consumes the household-level GPS coordinates that are already in the spine, (iii) links each pregnancy to a daily UTCI series from ERA5-Land at the household coordinate, and (iv) writes the pregnancy-level exposure matrix to `exposure.rds`. Per the workflow agreed with NHRC, all climate-variable generation — ERA5-Land download, UTCI computation following Di Napoli et

- al. (2018), and gestational-window aggregation — is conducted within this script and executed by the Centre’s analysts against the full extract.
3. Write `03_phase1_descriptive.Rmd`, `04_phase2_misclassification.Rmd`, `05_phase3_dlnm_perinatal.Rmd`, `06_phase4_casecrossover.Rmd`, `07_phase5_dlnm_birthweight.Rmd`, `08_phase6_effect_modification.Rmd`, and `09_phase7_sensitivity.Rmd`, in line with the Chapter 3 analytic plan. The effect-modification and ANC-related analyses use the 2013-onward `anc_reliable` sub-cohort; the primary perinatal-outcome models use the full birth-year range available in the NHRC extract.